

Unit 4: Key Types of Business Data Analytics: Descriptive, Predictive and Prescriptive

Assignment 9: Predictive and Prescriptive Analytics: A Comprehensive Study for Business Data Analytics using Python

Topics Covered: Predictive Analytics with Clustering (K-Means); Prescriptive Analytics with Decision Tree

The major objective of this assignment is to empower learners with the ability to extract meaningful insights from data and make informed decisions. Clustering helps in identifying patterns, segmenting markets, and detecting anomalies. Decision Trees assist in making interpretable and accurate predictions. This is useful for recommending suitable products or services to customers based on their characteristics and historical behavior. Optimization ensures efficient use of resources and strategic planning to achieve the best results under constraints.

Each technique complements different stages of data-driven decision-making processes.

Clustering:

Clustering plays a crucial role in business data analytics by allowing companies to segment their data, uncover trends, and make informed decisions that enhance customer experiences, improve operational efficiency, and ultimately drive business growth.

Recommendation System:

Recommendation system that utilizes the trained decision tree classifier to suggest suitable products or services to customers can be designed. Implement the recommendation system using Python.

Students are required to create a folder as FourDigitsRollno_Roomno_DeptCode_BDA (For Example, 0012_14_COMA_BDA) on their desktop. All the files related to the assignment should be saved within this folder. Students are required to use either Spyder or Jupyter Notebook to complete the assignment. The final working code should be saved as AssignmentNo_Rollno.py

Store Performance Segmentation using K-Means Clustering (Predictive Analytics)

1. A retail company operates multiple stores across different locations and wants to segment its stores based on performance to support decisions related to resource allocation, marketing focus, and expansion strategy. Store performance is evaluated using monthly sales revenue and customer footfall. Sales data for eight stores are provided below.

- Monthly Sales (\$): 1000, 1500, 800, 1200, 4000, 4200, 3900, 1000
- Number of Customers: 50, 60, 45, 55, 300, 320, 310, 52

To identify groups of similar-performing stores, apply K-Means clustering to divide the stores into two distinct clusters.

- a. Create a dictionary with two features: Monthly_Sales, Number_of_Customers
- b. Load the data into a Pandas DataFrame.
- c. Apply the K-Means clustering algorithm from Scikit-Learn to group the stores into two clusters.
- d. Fit the clustering model to the dataset.
- e. Append the generated cluster labels (0 or 1) to the DataFrame.
- f. Visualize the clusters using a scatter plot:
 - X-axis: Monthly Sales
 - Y-axis: Number of Customers
 - Use different colors for each cluster
 - Add an appropriate title and axis labels

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Business Data Analytics Practical under NEP

Decision Support System for Event Attendance Using Decision Tree (Prescriptive Analytics)

2. An event management company wants to help customers decide whether a comedy show is worth attending based on attributes of the comedian. A customer has maintained historical records of comedy shows, capturing details about the comedian and whether the show was attended or not. The dataset comedyshow.csv includes features such as comedian age, nationality, experience, comedy ranking, and the attendance decision.
- a. Load the dataset using the read_csv() function from the Pandas library.
 - b. Convert categorical variables into numerical format using suitable encoding techniques.
 - c. Define the independent variables and the dependent variable.
 - d. Split the dataset into training and testing sets, using 30% of the data for testing.
 - e. Initialize and train a Decision Tree Classifier.
 - f. Visualize the trained decision tree model.
 - g. Save the decision tree visualization as a .png file.
 - h. Use the trained Decision Tree as a prescriptive analytics tool to recommend whether the person should attend the following comedy show:
 - Comedian Age: 40 years
 - Nationality: American
 - Experience: 10 years
 - Comedy Rating: 7
 - i. Predict the outcome if the comedy rating is reduced to 6.
